# D01 | XOR decision block

All-case aligned observations: 12,536 | Train: 8,875 | Held-out test: 3,661

Test accuracy 0.789 | Majority baseline 0.789 | Balanced accuracy 0.500 | Macro-F1 0.441

B1 -> M01 | aligned support 2,609
B2 -> M07 | aligned support 9,927

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.AppType != "Limit raise" AND e.NumberOfTerms <= 41.5 AND c.RequestedAmount <= 10875 -> B2 [train n=1,109]

c.AppType != "Limit raise" AND e.NumberOfTerms <= 41.5 AND c.RequestedAmount > 10875 -> B2 [train n=369]

c.AppType != "Limit raise" AND e.NumberOfTerms > 41.5 AND c.LoanGoal != "Not speficied" -> B2 [train n=6,511]

c.AppType != "Limit raise" AND e.NumberOfTerms > 41.5 AND c.LoanGoal = "Not speficied" -> B2 [train n=399]

c.AppType = "Limit raise" AND e.OfferedAmount <= 8450 -> B2 [train n=59]

c.AppType = "Limit raise" AND e.OfferedAmount > 8450 AND e.OfferedAmount <= 10200 -> B2 [train n=60]

c.AppType = "Limit raise" AND e.OfferedAmount > 8450 AND e.OfferedAmount > 10200 -> B2 [train n=368]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D02 | XOR decision block

All-case aligned observations: 31,509 | Train: 22,056 | Held-out test: 9,453

Test accuracy 0.837 | Majority baseline 0.651 | Balanced accuracy 0.767 | Macro-F1 0.792

B1 -> A_Submitted | aligned support 20,423
B2 -> M03 | aligned support 11,086

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.AppType != "Limit raise" AND c.RequestedAmount <= 4500 -> B2 [train n=1,624]

c.AppType != "Limit raise" AND c.RequestedAmount > 4500 AND c.LoanGoal != "Unknown" -> B1 [train n=17,919]

c.AppType != "Limit raise" AND c.RequestedAmount > 4500 AND c.LoanGoal = "Unknown" -> B2 [train n=142]

c.AppType = "Limit raise" -> B2 [train n=2,371]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D03 | XOR decision block

All-case aligned observations: 31,509 | Train: 22,056 | Held-out test: 9,453

Test accuracy 0.787 | Majority baseline 0.694 | Balanced accuracy 0.830 | Macro-F1 0.777

B1 -> A_Cancelled | aligned support 9,927
B2 -> parallelGateway | aligned support 21,582

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 297 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 153.465 -> B1 [train n=3,103]

e.CreditScore <= 297 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 153.465 -> B1 [train n=7,618]

e.CreditScore <= 297 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 75 -> B2 [train n=1,500]

e.CreditScore <= 297 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 75 -> B2 [train n=541]

e.CreditScore > 297 -> B2 [train n=9,294]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D04 | XOR decision block

All-case aligned observations: 21,291 | Train: 14,806 | Held-out test: 6,485

Test accuracy 0.733 | Majority baseline 0.733 | Balanced accuracy 0.500 | Macro-F1 0.423

B1 -> M02 | aligned support 5,719

B2 -> A_Validating | aligned support 15,572

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.FirstWithdrawal <= 0.82499999 AND e.CreditScore <= 310 AND c.AppType != "New credit" -> B2 [train n=633]

e.FirstWithdrawal <= 0.82499999 AND e.CreditScore <= 310 AND c.AppType = "New credit" -> B2 [train n=1,003]

e.FirstWithdrawal <= 0.82499999 AND e.CreditScore > 310 AND c.LoanGoal != "Existing loan takeover" -> B2 [train
n=1,144]

e.FirstWithdrawal <= 0.82499999 AND e.CreditScore > 310 AND c.LoanGoal = "Existing loan takeover" -> B2 [train n=475]

e.FirstWithdrawal > 0.82499999 AND e.CreditScore <= 297 AND c.AppType != "New credit" -> B2 [train n=309]

e.FirstWithdrawal > 0.82499999 AND e.CreditScore <= 297 AND c.AppType = "New credit" -> B2 [train n=2,960]

e.FirstWithdrawal > 0.82499999 AND e.CreditScore > 297 AND e.NumberOfTerms <= 125.5 -> B2 [train n=7,915]

e.FirstWithdrawal > 0.82499999 AND e.CreditScore > 297 AND e.NumberOfTerms > 125.5 -> B2 [train n=367]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D05 | XOR decision block

All-case aligned observations: 20,423 | Train: 14,272 | Held-out test: 6,151

Test accuracy 0.829 | Majority baseline 0.829 | Balanced accuracy 0.500 | Macro-F1 0.453

B1 -> M05 | aligned support 16,930

B2 -> W_Handle leads | aligned support 3,493

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount <= 9550 AND c.LoanGoal != "Home improvement" -> B1 [train
n=3,129]

c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount <= 9550 AND c.LoanGoal = "Home improvement" -> B1 [train
n=1,065]

c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount > 9550 AND c.LoanGoal != "Not speficied" -> B1 [train
n=6,942]

c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount > 9550 AND c.LoanGoal = "Not speficied" -> B1 [train
n=460]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount <= 31727 AND c.RequestedAmount <= 20950 -> B1 [train
n=1,582]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount <= 31727 AND c.RequestedAmount > 20950 -> B1 [train n=544]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount > 31727 AND c.RequestedAmount <= 52750 -> B1 [train n=410]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount > 31727 AND c.RequestedAmount > 52750 -> B1 [train n=140]
```

Branches never predicted by this tree: B2

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D06 | XOR decision block

All-case aligned observations: 36,848 | Train: 25,807 | Held-out test: 11,041

Test accuracy 0.856 | Majority baseline 0.856 | Balanced accuracy 0.500 | Macro-F1 0.461

B1 -> M06 | aligned support 5,339

B2 -> parallelGateway | aligned support 31,509

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 597.5 AND c.AppType != "Limit raise" AND e.OfferedAmount <= 9650 -> B2 [train n=4,259]

e.CreditScore <= 597.5 AND c.AppType != "Limit raise" AND e.OfferedAmount > 9650 -> B2 [train n=9,818]

e.CreditScore <= 597.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 225 -> B2 [train n=1,655]

e.CreditScore <= 597.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 225 -> B2 [train n=549]

e.CreditScore > 597.5 AND c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 853.5 -> B2 [train n=2,784]

e.CreditScore > 597.5 AND c.LoanGoal != "Remaining debt home" AND e.CreditScore > 853.5 -> B2 [train n=6,436]

e.CreditScore > 597.5 AND c.LoanGoal = "Remaining debt home" AND e.NumberOfTerms <= 168 -> B2 [train n=237]

e.CreditScore > 597.5 AND c.LoanGoal = "Remaining debt home" AND e.NumberOfTerms > 168 -> B2 [train n=69]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D07 | XOR decision block

All-case aligned observations: 42,873 | Train: 29,823 | Held-out test: 13,050

Test accuracy 0.535 | Majority baseline 0.503 | Balanced accuracy 0.535 | Macro-F1 0.534

B1 -> D10 | aligned support 21,582
B2 -> A_Incomplete | aligned support 21,291

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.AppType != "New credit" AND c.LoanGoal != "Existing loan takeover" AND e.CreditScore <= 799 -> B1 [train n=2,424]

c.AppType != "New credit" AND c.LoanGoal != "Existing loan takeover" AND e.CreditScore > 799 -> B1 [train n=324]

c.AppType != "New credit" AND c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount <= 19750 -> B1 [train n=146]

c.AppType != "New credit" AND c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount > 19750 -> B1 [train n=125]

c.AppType = "New credit" AND c.LoanGoal != "Remaining debt home" AND e.MonthlyCost <= 200.07 -> B1 [train n=10,181]

c.AppType = "New credit" AND c.LoanGoal != "Remaining debt home" AND e.MonthlyCost > 200.07 -> B2 [train n=15,353]

c.AppType = "New credit" AND c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 10875 -> B2 [train n=871]

c.AppType = "New credit" AND c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 10875 -> B2 [train n=399]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D08 | XOR decision block

All-case aligned observations: 21,291 | Train: 14,806 | Held-out test: 6,485

Test accuracy 0.887 | Majority baseline 0.887 | Balanced accuracy 0.500 | Macro-F1 0.470

B1 -> M04 | aligned support 18,939
B2 -> W_Call incomplete files | aligned support 2,352

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.FirstWithdrawal <= 1918.475 AND e.CreditScore <= 767.5 AND e.CreditScore <= 716.5 -> B1 [train n=2,031]

e.FirstWithdrawal <= 1918.475 AND e.CreditScore <= 767.5 AND e.CreditScore > 716.5 -> B1 [train n=209]

e.FirstWithdrawal <= 1918.475 AND e.CreditScore > 767.5 AND c.RequestedAmount <= 49850 -> B1 [train n=2,239]

e.FirstWithdrawal <= 1918.475 AND e.CreditScore > 767.5 AND c.RequestedAmount > 49850 -> B1 [train n=136]

e.FirstWithdrawal > 1918.475 AND e.FirstWithdrawal <= 27900 AND e.NumberOfTerms <= 49.5 -> B1 [train n=2,157]

e.FirstWithdrawal > 1918.475 AND e.FirstWithdrawal <= 27900 AND e.NumberOfTerms > 49.5 -> B1 [train n=7,174]

e.FirstWithdrawal > 1918.475 AND e.FirstWithdrawal > 27900 AND c.LoanGoal != "Remaining debt home" -> B1 [train n=807]

e.FirstWithdrawal > 1918.475 AND e.FirstWithdrawal > 27900 AND c.LoanGoal = "Remaining debt home" -> B1 [train n=53]
```

Branches never predicted by this tree: B2

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D09 | XOR decision block

All-case aligned observations: 4,703 | Train: 3,255 | Held-out test: 1,448

Test accuracy 0.794 | Majority baseline 0.794 | Balanced accuracy 0.500 | Macro-F1 0.442

B1 -> M13 | aligned support 3,728
B2 -> M11 | aligned support 975

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.OfferedAmount <= 11950 AND e.FirstWithdrawal <= 1240 AND c.RequestedAmount <= 5950 -> B1 [train n=163]

e.OfferedAmount <= 11950 AND e.FirstWithdrawal <= 1240 AND c.RequestedAmount > 5950 -> B1 [train n=146]

e.OfferedAmount <= 11950 AND e.FirstWithdrawal > 1240 AND e.FirstWithdrawal <= 8725 -> B1 [train n=919]

e.OfferedAmount <= 11950 AND e.FirstWithdrawal > 1240 AND e.FirstWithdrawal > 8725 -> B1 [train n=244]

e.OfferedAmount > 11950 AND c.AppType != "Limit raise" AND c.RequestedAmount <= 11400 -> B1 [train n=227]

e.OfferedAmount > 11950 AND c.AppType != "Limit raise" AND c.RequestedAmount > 11400 -> B1 [train n=1,411]

e.OfferedAmount > 11950 AND c.AppType = "Limit raise" AND e.OfferedAmount <= 21250 -> B1 [train n=59]

e.OfferedAmount > 11950 AND c.AppType = "Limit raise" AND e.OfferedAmount > 21250 -> B1 [train n=86]
```

Branches never predicted by this tree: B2

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D10 | XOR decision block

All-case aligned observations: 21,582 | Train: 15,017 | Held-out test: 6,565

Test accuracy 0.876 | Majority baseline 0.825 | Balanced accuracy 0.904 | Macro-F1 0.824

B1 -> O_Accepted | aligned support 17,854
B2 -> A_Denied | aligned support 3,728

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 297 AND c.AppType != "New credit" AND e.NumberOfTerms <= 104.5 -> B1 [train n=620]

e.CreditScore <= 297 AND c.AppType != "New credit" AND e.NumberOfTerms > 104.5 -> B1 [train n=966]

e.CreditScore <= 297 AND c.AppType = "New credit" AND e.OfferedAmount <= 8950 -> B2 [train n=1,180]

e.CreditScore <= 297 AND c.AppType = "New credit" AND e.OfferedAmount > 8950 -> B2 [train n=2,957]

e.CreditScore > 297 -> B1 [train n=9,294]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D11 | XOR decision block

All-case aligned observations: 41,407 | Train: 29,001 | Held-out test: 12,406

Test accuracy 0.890 | Majority baseline 0.890 | Balanced accuracy 0.500 | Macro-F1 0.471

B1 -> M09 | aligned support 4,559
B2 -> M12 | aligned support 36,848

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 594.5 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 20.825 -> B2 [train n=4,971]

e.CreditScore <= 594.5 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 20.825 -> B2 [train n=11,837]

e.CreditScore <= 594.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 1508 -> B2 [train n=1,818]

e.CreditScore <= 594.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 1508 -> B2 [train n=441]

e.CreditScore > 594.5 AND c.AppType != "New credit" AND c.RequestedAmount <= 2650 -> B2 [train n=89]

e.CreditScore > 594.5 AND c.AppType != "New credit" AND c.RequestedAmount > 2650 -> B2 [train n=249]

e.CreditScore > 594.5 AND c.AppType = "New credit" AND c.LoanGoal != "Remaining debt home" -> B2 [train n=9,267]

e.CreditScore > 594.5 AND c.AppType = "New credit" AND c.LoanGoal = "Remaining debt home" -> B2 [train n=329]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D12 | XOR decision block

All-case aligned observations: 39,979 | Train: 28,007 | Held-out test: 11,972

Test accuracy 0.922 | Majority baseline 0.922 | Balanced accuracy 0.500 | Macro-F1 0.480

B1 -> M12 | aligned support 3,131
B2 -> D13 | aligned support 36,848

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 605 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 0.32499999 -> B2 [train n=4,637]

e.CreditScore <= 605 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 0.32499999 -> B2 [train n=11,279]

e.CreditScore <= 605 AND c.AppType = "Limit raise" AND e.OfferedAmount <= 13550 -> B2 [train n=531]

e.CreditScore <= 605 AND c.AppType = "Limit raise" AND e.OfferedAmount > 13550 -> B2 [train n=1,690]

e.CreditScore > 605 AND e.FirstWithdrawal <= 9692.8848 AND c.LoanGoal != "Remaining debt home" -> B2 [train n=5,850]

e.CreditScore > 605 AND e.FirstWithdrawal <= 9692.8848 AND c.LoanGoal = "Remaining debt home" -> B2 [train n=203]

e.CreditScore > 605 AND e.FirstWithdrawal > 9692.8848 AND e.MonthlyCost <= 246.635 -> B2 [train n=1,430]

e.CreditScore > 605 AND e.FirstWithdrawal > 9692.8848 AND e.MonthlyCost > 246.635 -> B2 [train n=2,387]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D13 | XOR decision block

All-case aligned observations: 36,848 | Train: 25,807 | Held-out test: 11,041

Test accuracy 0.542 | Majority baseline 0.509 | Balanced accuracy 0.544 | Macro-F1 0.535

B1 -> M10 | aligned support 17,919

B2 -> W_Complete application | aligned support 18,929

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.RequestedAmount <= 4500 AND c.AppType != "Limit raise" AND e.OfferedAmount <= 10650 -> B2 [train n=564]

c.RequestedAmount <= 4500 AND c.AppType != "Limit raise" AND e.OfferedAmount > 10650 -> B2 [train n=1,359]

c.RequestedAmount <= 4500 AND c.AppType = "Limit raise" AND e.OfferedAmount <= 12250 -> B2 [train n=113]

c.RequestedAmount <= 4500 AND c.AppType = "Limit raise" AND e.OfferedAmount > 12250 -> B2 [train n=446]

c.RequestedAmount > 4500 AND e.FirstWithdrawal <= 10697 AND e.FirstWithdrawal <= 194 -> B2 [train n=6,904]

c.RequestedAmount > 4500 AND e.FirstWithdrawal <= 10697 AND e.FirstWithdrawal > 194 -> B1 [train n=9,849]

c.RequestedAmount > 4500 AND e.FirstWithdrawal > 10697 AND c.RequestedAmount <= 10775 -> B1 [train n=139]

c.RequestedAmount > 4500 AND e.FirstWithdrawal > 10697 AND c.RequestedAmount > 10775 -> B1 [train n=6,433]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D14 | XOR decision block

All-case aligned observations: 21,582 | Train: 15,017 | Held-out test: 6,565

Test accuracy 0.615 | Majority baseline 0.509 | Balanced accuracy 0.617 | Macro-F1 0.613

B1 -> M17 | aligned support 10,857
B2 -> M16 | aligned support 10,725

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 852.5 AND e.OfferedAmount <= 26050 AND c.AppType != "Limit raise" -> B2 [train n=5,431]

e.CreditScore <= 852.5 AND e.OfferedAmount <= 26050 AND c.AppType = "Limit raise" -> B1 [train n=1,182]

e.CreditScore <= 852.5 AND e.OfferedAmount > 26050 AND e.CreditScore <= 730.5 -> B2 [train n=1,403]

e.CreditScore <= 852.5 AND e.OfferedAmount > 26050 AND e.CreditScore > 730.5 -> B2 [train n=455]

e.CreditScore > 852.5 AND e.OfferedAmount <= 25050 AND c.LoanGoal != "Remaining debt home" -> B1 [train n=5,148]

e.CreditScore > 852.5 AND e.OfferedAmount <= 25050 AND c.LoanGoal = "Remaining debt home" -> B2 [train n=134]

e.CreditScore > 852.5 AND e.OfferedAmount > 25050 AND e.FirstWithdrawal <= 2836 -> B2 [train n=445]

e.CreditScore > 852.5 AND e.OfferedAmount > 25050 AND e.FirstWithdrawal > 2836 -> B2 [train n=819]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D15 | XOR decision block

All-case aligned observations: 6,381 | Train: 4,466 | Held-out test: 1,915

Test accuracy 0.762 | Majority baseline 0.762 | Balanced accuracy 0.500 | Macro-F1 0.432

B1 -> M14 | aligned support 4,808
B2 -> M15 | aligned support 1,573

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.AppType != "New credit" AND e.NumberOfTerms <= 126.5 AND e.FirstWithdrawal <= 900 -> B1 [train n=124]

c.AppType != "New credit" AND e.NumberOfTerms <= 126.5 AND e.FirstWithdrawal > 900 -> B1 [train n=53]

c.AppType != "New credit" AND e.NumberOfTerms > 126.5 -> B1 [train n=51]

c.AppType = "New credit" AND e.OfferedAmount <= 6350 AND e.CreditScore <= 637 -> B1 [train n=206]

c.AppType = "New credit" AND e.OfferedAmount <= 6350 AND e.CreditScore > 637 -> B1 [train n=196]

c.AppType = "New credit" AND e.OfferedAmount > 6350 AND e.CreditScore <= 733.5 -> B1 [train n=2,042]

c.AppType = "New credit" AND e.OfferedAmount > 6350 AND e.CreditScore > 733.5 -> B1 [train n=1,794]
```

Branches never predicted by this tree: B2

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D16 | XOR decision block

All-case aligned observations: 17,854 | Train: 12,438 | Held-out test: 5,416

Test accuracy 0.837 | Majority baseline 0.731 | Balanced accuracy 0.716 | Macro-F1 0.749

B1 -> M14 | aligned support 13,046
B2 -> M15 | aligned support 4,808

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 596 AND c.AppType != "New credit" AND e.NumberOfTerms <= 129.5 -> B1 [train n=1,316]

e.CreditScore <= 596 AND c.AppType != "New credit" AND e.NumberOfTerms > 129.5 -> B1 [train n=119]

e.CreditScore <= 596 AND c.AppType = "New credit" AND e.OfferedAmount <= 8650 -> B2 [train n=322]

e.CreditScore <= 596 AND c.AppType = "New credit" AND e.OfferedAmount > 8650 -> B2 [train n=1,392]

e.CreditScore > 596 AND e.NumberOfTerms <= 60.5 AND e.OfferedAmount <= 5050 -> B1 [train n=797]

e.CreditScore > 596 AND e.NumberOfTerms <= 60.5 AND e.OfferedAmount > 5050 -> B1 [train n=2,974]

e.CreditScore > 596 AND e.NumberOfTerms > 60.5 AND c.AppType != "Limit raise" -> B1 [train n=5,232]

e.CreditScore > 596 AND e.NumberOfTerms > 60.5 AND c.AppType = "Limit raise" -> B1 [train n=286]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.

# D17 | XOR decision block

All-case aligned observations: 15,578 | Train: 10,957 | Held-out test: 4,621

Test accuracy 0.700 | Majority baseline 0.697 | Balanced accuracy 0.519 | Macro-F1 0.465

B1 -> M17 | aligned support 10,725

B2 -> M16 | aligned support 4,853

### Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 882.5 AND c.AppType != "Limit raise" -> B1 [train n=6,321]

c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 882.5 AND c.AppType = "Limit raise" -> B1 [train n=1,250]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 882.5 AND e.OfferedAmount <= 25800 -> B1 [train n=1,922]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 882.5 AND e.OfferedAmount > 25800 -> B1 [train n=825]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 10125 AND e.FirstWithdrawal <= 3051.535 -> B2 [train
n=346]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 10125 AND e.FirstWithdrawal > 3051.535 -> B2 [train n=103]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 10125 AND e.OfferedAmount <= 26000 -> B1 [train n=131]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 10125 AND e.OfferedAmount > 26000 -> B1 [train n=59]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 79.39% is measured without these guards.